CSA 0378-DATA STRUCTURES

22. DIKJSTRA’S ALGORITH

main.c

Share Run

```c
#include <stdio.h>

int main() {
    int graph[4][4] = {
        {0,1,4,0},
        {1,0,2,5},
        {4,2,0,1},
        {0,5,1,0}
    };

    int dist[4] = {0,999,999,999};
    int visited[4] = {0};
    int i,j,min,u;

    for(i=0;i<4;i++) {
        min = 999;

        for(j=0;j<4;j++)
            if(!visited[j] && dist[j] < min) {
                min = dist[j];
                u = j;
            }

        visited[u] = 1;

        for(j=0;j<4;j++)
            if(graph[u][j] && dist[u] + graph[u][j] < dist[j])
                dist[j] = dist[u] + graph[u][j];
    }

    printf("Distances:\n");
    for(i=0;i<4;i++)
        printf("%d ", dist[i]);

    return 0;
}
```

Output

```
Distances:
0 1 3 4

=== Code Execution Successful ===
```